

Computer-assisted Acute Lymphoblastic Leukemia detection and diagnosis

Sarmad Shafique

Dept of computer science

Bahria University

Islamabad, Pakistan

sarmadkashmiri05@gmail.com

Samabia Tehsin

Dept of computer science

Bahria University

Islamabad, Pakistan

stehseen.buic@bahria.edu.pk

Syed Anas

Dept of computer science

Bahria University

Islamabad, Pakistan

syedanasshah@outlook.com

Farrukh Masud

Dept of computer science

Bahria University

Islamabad, Pakistan

farrukhmasud221@gmail.com

Abstract—Leukemia is a cancer of white blood cells (WBCs) which damages blood and bone marrow of human body. It can be fatal disease if not diagnose at earlier stage. Generally complete blood count (CBC) or morphological image analysis is used to manually diagnose the leukemia cells. These methods are time consuming and less accurate which needs to be fixed. In this paper we have proposed an automated technique for the detection of acute lymphoblastic leukemia by microscopic blood image analysis. This approach first segment out the different types of cells from the image i.e. white blood cells, red blood cells and platelets. After that Lymphocytes are separated from the white blood cells. Then shape and color features are extracted from these lymphocytes which are given to SVM classifier to classify the cells into normal and blast. We have achieved an overall accuracy of 93.7%, sensitivity of 92% and specificity of 91% by testing our proposed algorithm over the ALL-IDB-1 dataset. This automated leukemia detection system found to be more effective, fast and accurate as compare to manual diagnosing methods.

Index Terms—acute leukemia detection, image processing, lymphocytes, medical image analysis.

I. INTRODUCTION

Acute lymphoblastic leukaemia is a type of cancer associated with blood in which leukocytes (WBCs) start growing abnormally. These abnormal cells strike the blood and bone marrow due to which immune system of human body become vulnerable. Furthermore, it suppresses the production of normal red blood cells and platelets hence lead towards anemia, the blood deficiency [1]. Moreover, these abnormal leukocytes predominantly spread into the human blood swiftly and can also capture other different body parts like kidney, liver, spleen, brain and lymph nodes.

Leukemia is classified as either Lymphoblastic or Myelogenous depending on the type of white blood cells being infected. If the infected cells are granulocytes and monocytes, then the leukemia will be classified as Myelogenous (AML) and if the infected cells are lymphocytes, then the leukemia will be classified as Lymphoblastic (ALL) [2]. According to French American British (FAB) classification, ALL is further categorized into 3 subtypes, which are L1, L2 and L3. L1 type cells are normally small in size and are ho-

mogeneous with little cytoplasm. Their nucleus is discoid and well structured. L2 type cells have shape dissimilarity and are over-sized as compared to L1. Their nucleus is not regular and contains variations in their cytoplasm. L3 type cells are of identical shape and normal size with round or oval nucleus. They have adequate amount of cytoplasm which includes vacuoles. They are usually larger in size than L1.

In this study we have only investigated ALL and our aim is to classify the lymphocytes into normal or blast cells. Generally leukemia is diagnosed by performing CBC (complete blood count) test in which hematologists check the entire count of white blood cells, red blood cells and platelets. Although, this CBC test may show some signs of leukemia but this is not adequate for the hematologists to diagnose the patient with leukemia. So, they have to utilize different methodologies including bone marrow aspiration [3] and microscopic examining of blood smear [4] to confirm the leukemia. However, all these manual methods demand a lot of endeavor and time to take place. Exceptionally trained medical professionals required to perform these examining, therefore it is labor intensive work. As contrary, automated leukemia detection systems can overcome the deficiency of manual diagnosing. Moreover, it will lessen the load of medical professionals and will provide the higher accuracy and speed as compare to manual diagnosing methods.

In this paper we proposed an automated leukemia detection technique by using microscopic blood samples. Our proposed method will first separate WBCs from the other blood cells and then lymphocytes will be extracted from these WBCs. After that different features including shape and color features are extracted and given to the support vector machine classifier for detection of normal and blast cells. Rest of the paper is organized as follows: Section 2 presents the previous related work. Section 3 explains the framework and methodology of the proposed system Section 4 presents the experimental results and analysis. In the end Section 5 presents the conclusion.

II. RELATED WORK

To address the manual diagnosing limitations, studies have presented different automated detection methods for ALL where microscopic blood image analysis is used to perform leukemia detection. These automated methods found to be more efficient, fast, cost effective and accurate as compared to manual methods. Karthikeyan et al [5] used histogram equalization and median filtering to perform pre-processing of blood images. Then fuzzy c-mean was carried out for the segmentation of white blood cells. After extracting features using Gabor texture extraction method, classification was carried out using support vector machine (SVM) to classify normal and blast cells. They had achieved 90% accuracy by their methodology. Y.Li et al [6] proposed a dual threshold method for the segmentation of lymphocytes, where they have achieved decent accuracy. They enhanced the standard single threshold method by utilizing golden section search to determine optimal threshold value of lymphocytes. MoradiAmin et al [7] performed ALL recognition by deploying fuzzy c mean clustering for lymphocytes segmentation. Principle component analysis used to reduce different shape based features. After that, SVM was trained over those features to classify the normal and blast lymphocytes. Putzu et al [8] used triangle threshold for segmentation of white blood cells. Features including shape features, colour features and texture features were extracted using GLCM. After that, SVM was used for classification of blast and normal cells. Mohapatra et al [9] used contrast enhancement and selective median filtering for preprocessing of the images. After that Shadowed c-mean clustering was used for the segmentation of lymphocytes into 3 regions i.e. background, cytoplasm and nucleus. After that, different features including fractal dimension, shape based features, colour features and texture features were extracted from those lymphocytes. Then an ensemble classifier (Naive Bayesian, K-nearest neighbor, Multilayer perceptron, Radial basis functional neural network, Support vector machines) was trained to classify normal and blast cells. There are lot of similar researches in literature for the WBC segmentation and leukemia detection. After inspecting the previous studies, we come to the point that there is still research room available to improve the accuracy of leukemia detection. ALL diagnosis is a highly sensitive issue which is related to the health and life of humans. Accuracy of such diagnoses systems should be unblemished for full replacement of human operators by these automated detection systems. But this is very challenging task because these blood cells are highly overlapping as shown in Figure 1. Which makes them difficult to separate. We proposed a robust and vigorous automated leukemia detection method to overcome above limitations.

III. METHODOLOGY

The process for automated leukemia detection consists of 5 major modules including preprocessing, segmentation, identification and separation of grouped lymphocytes, feature extraction and classification. Our methodology overview is

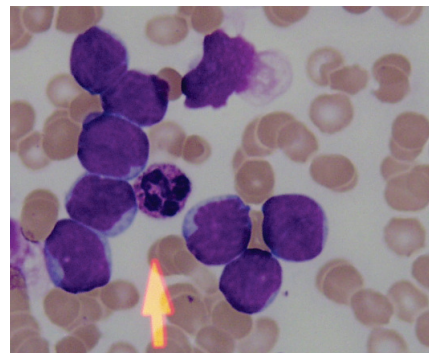


Fig. 1: Overlapping Cells in ALL-IDB-1

illustrated in Figure 2. A microscopic blood image comprises of white blood cells, red blood cells and platelets. The proposed architecture first perform preprocessing over these blood smears. Then segmentation is carried out to identify the lymphocytes. After that grouped lymphocytes are identified and separated. Then different features are extracted from the cells and classification is carried out to classify normal and blast cells.

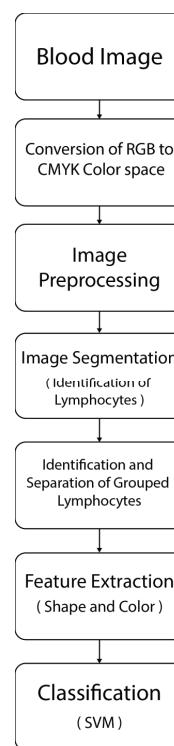


Fig. 2: Methodology Overview

A. Dataset

Blood images utilize in this research are acquired from ALL-IDB-1 dataset which is available online as a public dataset [10]. It consists of total 108 images where 59 images are from healthy patients and 49 images are affected with leukemia. All images have JPG format with 24 bit color depth

and a resolution of 2592x1944. These images have 3 types of blood cells present i-e Red blood cells, White blood cells and Platelets. To detect the leukemia we have to separate white blood cells from other components of cell. We have divided the dataset into training samples and testing samples. The training sample consists of 40 images whereas testing sample has 68 images which can be used for testing and evaluating the result. Blood sample images from ALL-IDB-1 dataset are shown in Figure 3.

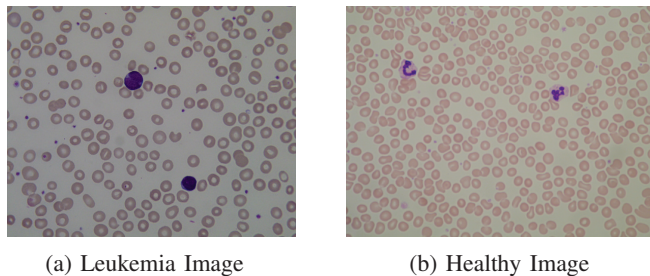


Fig. 3: Blood Sample Images from ALL-IDB-1

B. Preprocessing

In preprocessing we have first transformed the blood images from RGB (red green blue) to CMYK (cyan, magenta, yellow and black) color model. This highlights the WBCs from the other cells present in the blood image. Due to exposure of microscope, images have effects of unnecessary noise and blurriness. This may effect the quality of blood images and create chances of wrong diagnosis. Histogram equalization is a very common method that uses the image histogram to tune the contrast of image. [11] Therefore in preprocessing step we have applied histogram equalization over the blood image to overcome different lightning effects during image grabbing. Sample Images after preprocessing are shown in Figure 4.

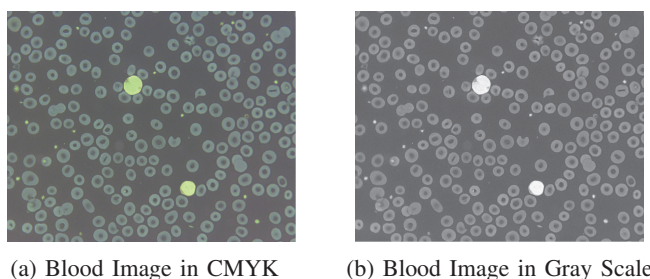


Fig. 4: Blood Sample Image during Preprocessing

C. Image Segmentation

Segmentation is a process of obtaining region of interest by dividing the image into different regions. For the detection of lymphocytes from the blood sample we have to separate WBCs from other cells and from the background. So we have applied Triangle oriented threshold method called Zack's Algorithm [12] over the blood samples to obtain required WBCs. In this algorithm a straight line is constructed between

highest value of the image histogram and the lowest value of image histogram. After this an optimal threshold value is calculated and segmentation is carried out using that threshold value. Zack algorithm is found to be more effectual when the histogram shows limpid difference between the maximum and minimum value.

After applying thresholding algorithm area opening function with the structuring element having circular shape is used to further clean the segmented image. After applying area opening all the objects which are smaller than the structuring element are removed from the image. The size of structuring element was calculated depending on the average size of different objects present in the blood image. Figure 5 shows the result of segmentation and area opening.

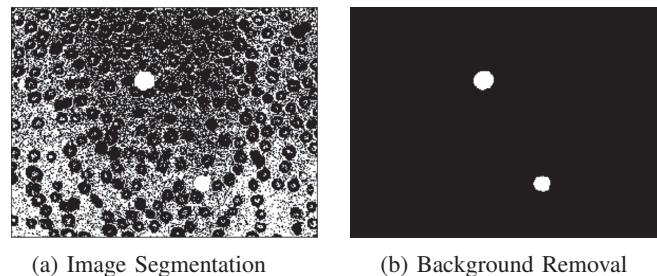


Fig. 5: Segmentation Result

D. Identification and Separation of Grouped Lymphocytes

In this stage we have to identify and separate the grouped and ungrouped lymphocytes from the image. For this purpose we have utilize roundness ratio to separate the grouped lymphocytes.

$$Roundness = \frac{4 * \pi * area}{(convex_perimeter)^2} \quad (1)$$

When roundness is equal to 1 it means that the detected object is circular and if the roundness is less than 1 it means the object is not circular. We perceived that a roundness ratio of 0.85 will be good to distinguish single lymphocyte from the group of lymphocytes. So the objects having roundness ratio greater then the provided threshold are classified as single lymphocyte and the objects having roundness ratio less than the provided threshold are classified as grouped lymphocytes which needs to be separated in the next step. The image containing grouped lymphocytes is shown in Figure 6.

To separate the grouped lymphocytes we have applied watershed segmentation to the distance transform. It is a image segmentation method which starts from the initial pixel called marker and consistently torrent the all other adjoining pixels of that marker known as catchment basin. The advantage of this technique over other methods is that the regions develop from the resulting boundaries are closed and connected as compared to standard edge based methods that create disconnected boundaries and usually post-process is required to form closed regions [13]. To remove the unnecessary spots and to fill the small holes in the nucleus of the cells binary morphological

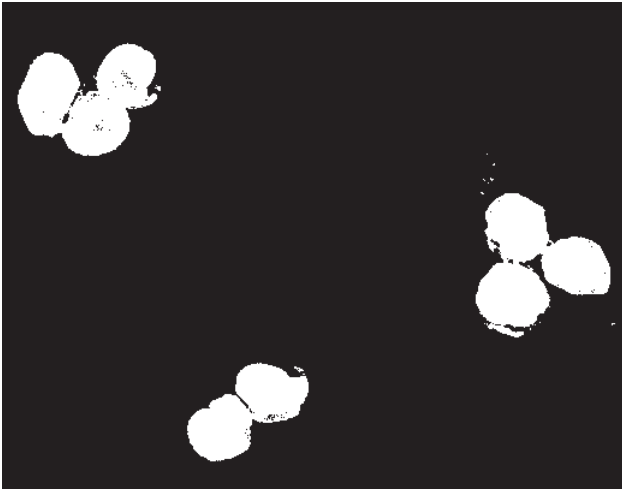


Fig. 6: Grouped Lymphocytes

opening and closing operation are performed over the blood image. Size of structuring element used in morphological operation is smaller than the nucleus but have enough size to remove the unnecessary spots and fill the small holes. Final resulted images are shown in Figure 7

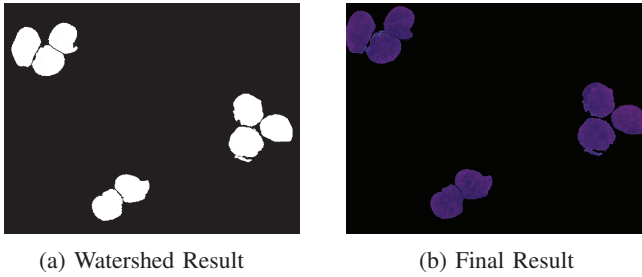


Fig. 7: Result of Watershed and Morphological Operations

E. Features Extraction

In features extraction we retrieved the information from the images like their specific structure of shape, color, texture etc. By using features extraction we can reduce the dimensions of images in such a way that they become more instructive and less redundant. This method is very useful when an algorithm has to process huge dataset which can be repetitive also, then by applying this method the dataset will be reduced to minimum dimensions which contains the most relevant information of the image. This data will be given to algorithm as input which will be easier for the algorithm to interpret and compute the results.

For leukemia detection we have extracted shape and color features to obtain the relevant information from the blood images. We can extract the shape features by using area, perimeter, convex perimeter, convex area, major and minor axis and orientation. To get all these calculations we have to compute Elongation, Eccentricity, Solidity and Convexity. Below are the formulas used to compute these shape based features.

$$Area = 2 * \pi * radius \quad (2)$$

$$Solidity = \frac{Area}{Convex_Area} \quad (3)$$

$$Elongation = 1 - \frac{minoraxis}{majoraxis} \quad (4)$$

$$Eccentricity = \frac{\sqrt{(majoraxis^2 - minoraxis^2)}}{majoraxis} \quad (5)$$

$$Convexity = \frac{perimeter_{convex}}{perimeter} \quad (6)$$

These shape based features can vulnerable to errors occur during segmentation process. Therefore, these features are used together with color features which are less vulnerable. The color features we have used are mean, variance, standard deviation and kurtosis. All these features are used as input feature vector for the classifier to perform effectively for the detection of leukemia. Below are the formulas used to compute these features.

$$Mean = \frac{\sum x}{n} \quad (7)$$

$$Variance = \frac{\sum (x - mean)^2}{n - 1} \quad (8)$$

$$StandardDeviation = \sqrt{Variance} \quad (9)$$

$$Kurtosis = \frac{\frac{\sum (x - mean)^4}{n - 1}}{(\sqrt{Variance})^4} \quad (10)$$

F. Classification

For classification we have used Support Vector Machine (SVM) classifier which is a supervised learning algorithm. It is very effective when we have to perform binary classification therefore we have used this algorithm because we have to classify two classes normal and blast. We have given our extracted feature vector as an input to the SVM which then classify the cells into normal and blast depending on their respective properties. SVM is used with the linear kernel to achieve the maximum accuracy and is effective in the high dimensional spaces. When the number of features are extensive, other kernel may not be able to map with high performance. Linear kernel is much faster, more accurate and used in order to produce slightly different decision boundaries [14]. It is seen that SVM perform effectively for the detection of ALL on the given feature vector. SVM classifier is illustrated in Figure 8.

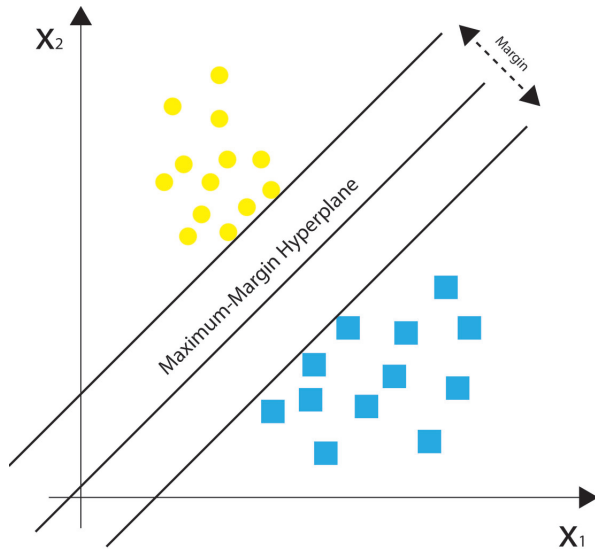


Fig. 8: SVM Classifier

IV. EXPERIMENTAL RESULTS AND ANALYSIS

We have validated our proposed algorithm over the ALL-IDB-1 dataset which consists of 108 blood images. These images are in jpg format and have resolution of 2592x1944. The test was performed on 68 blood images which were kept for testing purpose from the dataset. Overall accuracy of 93.70%, sensitivity of 92% and specificity of 91% was attained. It is seen that SVM perform effectively with the linear kernel and able to achieve a decent accuracy. Table I shows the result of some sample blood images tested with the proposed algorithm. It is seen that out of 89 cells, 84 were successfully detected by our automated system. We have compared our results with previous state-of-art classifiers and it is seen that our proposed methodology performed well as compared to previous methods. Putzu et al. [15] used the same dataset with support vector machine classifier and they were able to achieve an accuracy of 92%. Similarly Chatap et al. [16] used the same dataset and applied K-nearest neighbor (KNN) classifier for classification of normal and blast cells. They were able to achieve 93% accuracy with their methodology. Joshi et al. [17] also applied K-nearest neighbor (KNN) classifier on Public ALL-IDB 1 dataset and they were able to achieve 93% accuracy from their proposed methodology. We have applied Support vector machine over the same Public ALL-IDB 1 dataset. With effective preprocessing and segmentation techniques we were able to achieve higher accuracy of 93.70%. Table II shows the comparison with state-of-art methodologies.

V. CONCLUSION

In this paper, efforts have been made for the detection of acute lymphoblastic leukemia from microscopic blood images by using image processing techniques. Preprocessing was applied over the images to remove any noise, then segmentation is performed to detect lymphocytes from the image. Watershed is used to separate the grouped lymphocytes,

TABLE I: Performance of Proposed Algorithm

Image	Manual Count	Auto Count	Accuracy
Im020_1	02	02	100%
Im022_1	04	03	75%
Im024_1	04	04	100%
Im061_1	26	26	100%
Im012_1	14	13	92%
Im051_1	17	15	88%
Im008_1	18	17	94%
Im034_0	0	0	100%
Im035_0	1	1	100%
Im046_0	0	0	100%
Im097_0	2	2	100%
Im101_0	1	1	100%
Total Images 12	89	84	95.75%

TABLE II: Comparison with State-of-the-Art Classifiers

Method	Dataset	Classifiers	Accuracy %
Karthikeyan et al. 2017 [5]	Google (19 images)	Support Machine	90.00
Putzu et al. 2013 [8]	Public ALL-IDB 1 (108 images)	Support Machine	92.00
Putzua et al. 2014 [15]	Public ALL-IDB 1 (108 images) , ALL-IDB 2 (260 images)	Support Machine	92.00
Chatap et al. 2014 [16]	Public ALL-IDB 1 (108 images) , ALL-IDB 2 (260 images)	K-nearest neighbor (KNN)	93.00
Joshi et al. 2013 [17]	Public ALL-IDB 1 (108 images)	K-nearest neighbor (KNN)	93.00
Proposed	Public ALL-IDB 1 (108 images)	Support Vector Machine	93.70

after extracting shape and color features, SVM is used to classify normal and blast cells. We have achieved an overall accuracy of 93.70% by testing our proposed algorithm over the ALL-IDB-1 dataset. In future, we can further improve this system to detect different types of leukemia and other blood related diseases. Also we can further classify the leukemia

into its subtypes from the FAB classification which are L1, L2 and L3. Another direction for future work is to apply deep learning algorithm for detection which can automatically detect different visual feature from the images with no need of segmentation. But for this purpose we have to increase the dataset to train deep neural networks.

VI. CONFLICT OF INTEREST

The authors confirm that this article content has no conflict of interest.

VII. ACKNOWLEDGMENT

Declared None.

REFERENCES

- [1] Hematology.org, 'Acute Lymphocytic Leukemia: Learning Objectives', 2017. [Online]. Available: <http://www.hematology.org/Educators/Learning-Objectives/690.aspx>. [Accessed: 23- may- 2018].
- [2] Dana-farber.org, 'What Are the Differences Between Lymphocytic and Myelogenous Leukemia?', 2016.[Online]. Available: <https://blog.dana-farber.org/insight/2016/07/what-are-the-differences-between-lymphocytic-and-myelogenous-leukemia>. [Accessed: 23- may- 2018].
- [3] Curesearch.org, ' Acute Lymphoblastic Leukemia (ALL) in Children – In Treatment', 2010.[Online]. Available: <https://curesearch.org/Acute-Lymphoblastic-Leukemia-In-Treatment>. [Accessed: 28- may- 2018].
- [4] Cancer.org, 'Treating Acute Lymphocytic Leukemia (ALL)', 2007.[Online]. Available: <https://www.cancer.org/cancer/acute-lymphocytic-leukemia/treating.html>. [Accessed: 28- may- 2018].
- [5] Karthikeyan.T, and N. Poornima. "Microscopic image segmentation using Fuzzy C means for leukemia diagnosis." *Leukemia* 4, no. 1 (2017).
- [6] Li, Yan, Rui Zhu, Lei Mi, Yihui Cao, and Di Yao. "Segmentation of white blood cell from acute lymphoblastic leukemia images using dual-threshold method." *Computational and mathematical methods in medicine* 2016 (2016).
- [7] MoradiAmin, Morteza, Nasser Samadzadehaghdam, Saeed Kermani, and Ardeshtir Talebi. "Enhanced recognition of acute lymphoblastic leukemia cells in microscopic images based on feature reduction using principle component analysis." *Frontiers in Biomedical Technologies* 2, no. 3 (2015): 128-136.
- [8] Putzu, Lorenzo, and Cecilia Di Ruberto. "White blood cells identification and counting from microscopic blood image." In *Proceedings of World Academy of Science, Engineering and Technology*, no. 73, p. 363. World Academy of Science, Engineering and Technology (WASET), 2013.
- [9] Mohapatra, Subrajeet, Dipti Patra, and Sanghamitra Satpathy. "An ensemble classifier system for early diagnosis of acute lymphoblastic leukemia in blood microscopic images." *Neural Computing and Applications* 24, no. 7-8 (2014): 1887-1904.
- [10] Labati, Ruggero Donida, Vincenzo Piuri, and Fabio Scotti. "All-IDB: The acute lymphoblastic leukemia image database for image processing." In *Image processing (ICIP), 2011 18th IEEE international conference on*, pp. 2045-2048. IEEE, 2011.
- [11] Wang, Qian, Liya Chen, and Dinggang Shen. "Fast histogram equalization for medical image enhancement." In *Engineering in Medicine and Biology Society, 2008. EMBS 2008. 30th Annual International Conference of the IEEE*, pp. 2217-2220. IEEE, 2008.
- [12] Zack, G. W., W. E. Rogers, and S. A. Latt. "Automatic measurement of sister chromatid exchange frequency." *Journal of Histochemistry & Cytochemistry* 25, no. 7 (1977): 741-753.
- [13] Shafique, Sarmad, and Samabia Tehsin. "Computer-Aided Diagnosis of Acute Lymphoblastic Leukaemia." *Computational and mathematical methods in medicine* 2018 (2018).
- [14] Hsu, Chih-Wei, Chih-Chung Chang, and Chih-Jen Lin. "A practical guide to support vector classification." (2003): 1-16.
- [15] Putzu, Lorenzo, Giovanni Caocci, and Cecilia Di Ruberto. "Leucocyte classification for leukaemia detection using image processing techniques." *Artificial intelligence in medicine* 62, no. 3 (2014): 179-191.
- [16] Chatap, Niranjana, and Sini Shibu. "Analysis of blood samples for counting leukemia cells using Support vector machine and nearest neighbour." *IOSR Journal of Computer Engineering (IOSR-JCE)* 16, no. 5 (2014): 79-87.
- [17] Joshi, Minal D., Atul H. Karode, and S. R. Suralkar. "White blood cells segmentation and classification to detect acute leukemia." *International Journal of Emerging Trends & Technology in Computer Science (IJETTCS)* 2, no. 3 (2013): 147-151.